

BioMTAN: A Biological Knowledge-Guided Multi-Task Attention Network for Co-Enhanced Cancer Diagnosis and Prognosis

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Abstract—With the advancement of precision medicine, gene expression data have become a crucial tool in both cancer diagnosis and prognosis for different cancer types. The incorporation of biological pathways as prior knowledge has gained increasing interest in tackling the difficulties of high dimensionality and noisy information within gene expression data. However, most existing approaches guided by biological pathways ignore the intrinsic link between diagnostic and prognostic tasks in cancer research. They fail to capitalize on the potential of leveraging shared biological information from both tasks to enhance gene pathway representations. To this end, we introduce the Biological Knowledge-guided Multi-task Attention Network (BioMTAN), a novel multi-task learning framework designed for simultaneous prediction of molecular subtypes and survival risk. Specifically, we compile tailored knowledge collections that comprise multiple pathways for the two tasks, model them as unique subgraphs and use a multi-level information fusion strategy to provide a wealth of biological insights. Moreover, we develop a Multi-task Attention Module, which extracts essential global information functioning as the key and value by interacting with biological pathways from different collections, and utilizes task-specific local information as the query, efficiently decoding task-awareness feature for each task and facilitating communication across tasks within cancer diagnosis and prognosis. Extensive

validation on the public The Cancer Genome Atlas (TCGA) datasets confirms the enhanced performance of BioMTAN and highlights the significant pathways in each task, underscoring its potential as an instrumental asset in precision oncology.

Index Terms—Gene expression data, prior biological knowledge, multi-task learning.

I. INTRODUCTION

PRECISION medicine, crucial for providing personalized treatment with improved outcomes and reduced costs [1], has increasingly focused on cancer subtyping and prognostic assessment to advance precision oncology [2]. Gene expression profiles, adept at identifying individual intrinsic changes, play a key role in yielding biological insights for precision treatment [3], [4]. For example, Jubair et al. [5] introduced a network-based method that combines protein-protein interaction (PPI) networks with gene expression data to identify network biomarkers for breast cancer survivability and predict patient outcomes based on breast cancer subtypes. Zhou et al. [6] developed a deep learning model that utilizes multi-omics data, including gene expression, to generate 2D embeddings of high-dimensional features and predict Nottingham Prognostic Index (NPI) score levels. However, challenges arise due to the high dimensionality and noise inherent in gene expression data [7], as well as the limited interpretability of models. These hinder the broader application of gene expression data in routine clinical practice.

In contrast to singular genes, mapping gene expression data onto biological pathways with prior knowledge can be a strategic method to efficiently reduce the feature space and extract meaningful genes. Biological pathways are organized collections of genes and their interactions [8], collaboratively executing specific biological functions. Moreover, biological pathways offer a more comprehensive insight into disease conditions and biological mechanisms, capturing an integrated perspective of molecular activities within an organism [9]. Consequently, some researchers focused on utilizing biological pathways for cancer-related predictive modeling and achieved notable success recently. Lee et al. [10] modeled each pathway by a graph convolutional network and used a multi-attention-based ensemble model to combine several hundreds of pathways to classify cancer types. Luo Yuan [11] proposed the subhypergraph inductive

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neural network (SHINE) with thousands of pathways, which uses informative genetic pathways as hyperedges to connect genes as nodes for cancer type prediction. Wang et al. [12] developed GraphSurv which incorporated the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway association relationships between genes as a graph and employed a graph convolutional network to condense multi-omics data for cancer prognosis analysis.

Although these biological pathways-guided methods demonstrated promising results in cancer diagnosis and prognosis tasks, most methods were developed independently for each task. Consequently, they miss the potential of leveraging the biological interconnection of the two tasks, which may derive a better representation of the underlying biological processes of cancer and ultimately lead to better prediction in both tasks. In this context, multi-task learning stands out as a robust framework that improves learning efficiency and predictive precision by harnessing similarities and differences across tasks [13]. Nevertheless, employing genetic data for multi-task prediction can inadvertently introduce genetic information irrelevant to specific tasks (i.e., noise) easily. The challenge lies in disentangling task-specific information, while simultaneously leveraging shared biological insights from related tasks to enhance predictive accuracy in cancer diagnosis and prognosis.

To overcome these limitations, we propose a novel multi-task learning framework termed **Biological Knowledge-guided Multi-task Attention Network (BioMTAN)** to predict molecular subtypes of cancers and survival risks. Recognizing that critical genes and biological pathways differ across various disease-related tasks [14], [15], our approach integrates existing biological knowledge to create tailored pathway collections for each subtask. To capture the representation of each pathway effectively, we represent each pathway as a subgraph to extract and fuse multi-level features through Multi-level Graph Learning Module for the pathway's complete representation. Moreover, we introduce a Multi-task Attention Learning Module, comprising the Global Learning Block and the Task-driven Learning Block. Global Learning Block enhances the interaction among different biological pathways from different collections and extract pivotal global information. The Task-driven Learning Block uses task-specific local information (diagnostic or prognostic information) as the query and global information as the key and value to decode the task awareness feature for each prediction task. BioMTAN provides meaningful biological information from related task and focuses on task awareness features. Extensive experiments on three public TCGA datasets demonstrate the effectiveness and superiority of our model, showing mutual enhancement in cancer diagnosis and prognosis. We also highlight the significant pathways in each task to provide model interpretation, offering insights for targeted therapies and personalized medicine.

Our contributions can be summarized as follows:

- We propose a novel Biological Knowledge-guided Multi-task Attention Network (BioMTAN) that uniquely integrates biological pathway knowledge to predict molecular cancer subtypes and survival risks, facilitating mutual

enhancement in cancer diagnosis and prognosis through a multi-task paradigm.

- We strategically curate task-specific collections of biological pathways, enabling BioMTAN to extract and fuse multi-level biological insights. This explicit integration of tailored biological knowledge enhances predictive accuracy and provides meaningful interpretability.
- We design a Multi-task Attention Learning Module that enhances interactions among biological pathways and between various tasks. This module efficiently extracts essential global information from the interactions within biological pathways. It employs task-specific local information as the query and global information as both the key and value, effectively decoding task-awareness features for each task and learning richer information from related task.

The remained content of this paper is organized as follows. Related work is briefly introduced in Section II. In Section III, the details of the proposed methodology is described, which mainly includes biological knowledge-guided graph construction, the proposed Multi-level Graph Learning module, and Multi-task Attention module. Experimental results and analysis are illustrated in Section IV. Finally, Section V concludes this paper.

II. RELATED WORK

A. Gene Expression Data Analysis

In recent years, researchers have dedicated significant efforts to analyzing diseases using gene expression data. Some works concentrated on extracting highly variable genes (HVG) to make disease prediction [16], [17], [18], [19], [20], [21]. Qu et al. [17] used the Harris Hawks optimizer for gene feature extraction and a naive Bayesian classifier for classification. Shen et al. [18] proposed a deep learning approach combining a convolutional neural network (CNN) with a bidirectional gated recurrent unit (BiGRU) to achieve nonlinear dimensionality reduction and effectively filter out irrelevant factors in gene expression data. Zhao et al. [19] introduced CUP-AI-Dx, a classifier based on Ribonucleic Acid (RNA) that utilizes a 1D Inception CNN model trained on gene expression data, designed to identify a tumor's primary tissue of origin with high accuracy. These methods heavily depend on the dimensionality reduction of high-dimensional genes and the extraction of effective genes.

Recent studies explored the topological information contained in gene networks based on existing biological pathways and introduced graph neural network (GNN) to build models, showing excellent performance. Xing et al. [22] processed omics data into co-expression graphs using weighted correlation network analysis and utilized graph neural networks to enhance disease diagnosis and prognosis. Tan et al. [23] used a hierarchical graph convolution network (HiGCN) to learn better representations of gene expression data. They captured the relationships among genes while overlooked the intricate interplay of interactions within the biological pathways. Additionally, some studies used Transformer-based models to further improve the performance of gene expression data analysis.

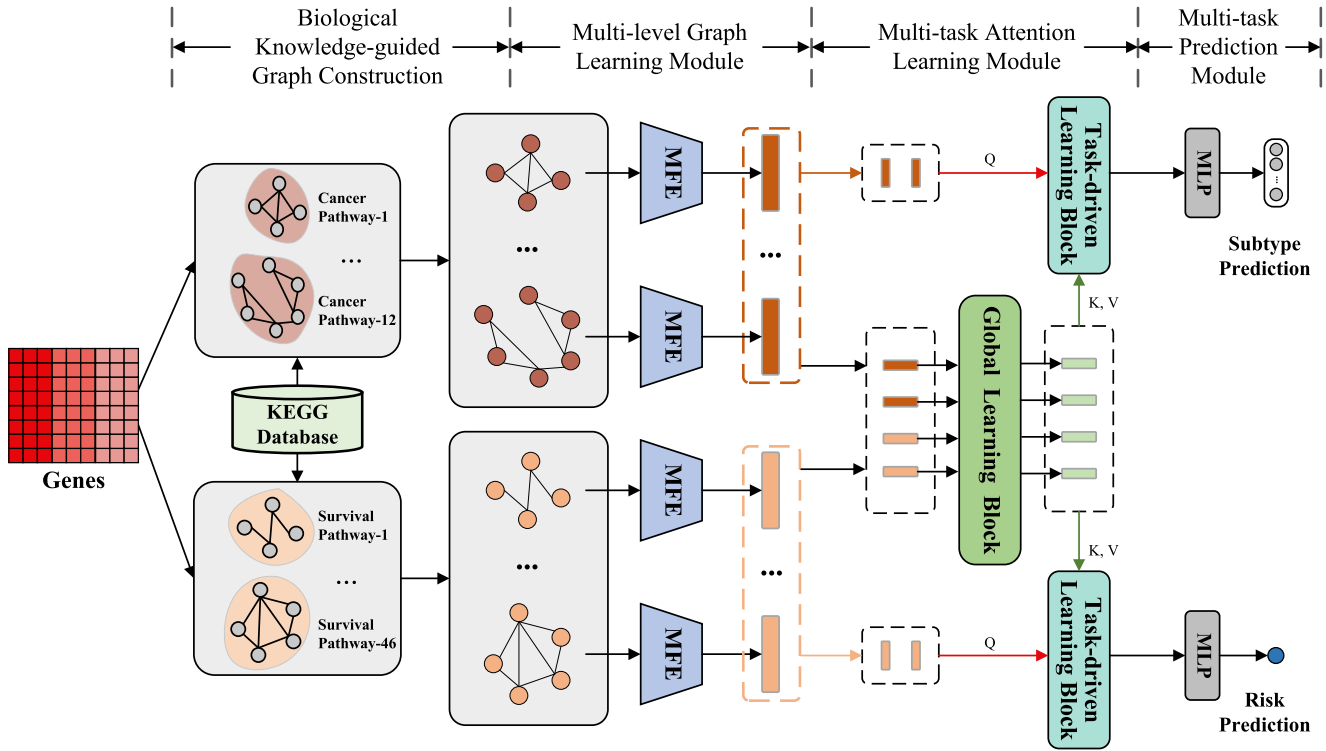


Fig. 1. Architecture of our proposed BioMTAN. First, graph sets are constructed using gene expression data and biological pathway collections. Next, the Multi-level Feature Extractor (MFE) in the Multi-level Graph Learning Module extracts biological pathway feature representations. These representations are then processed in the Multi-task Attention Module, where the Global Learning (GL) Block generates global key information, and the Task-driven Learning (TL) Block uses task-specific local information as the query (Q) and global information as the key (K) and value (V) to decode task-aware features. Finally, the Multi-task Prediction Module, implemented as Multi-layer Perceptron (MLP), outputs the predicted subtype and risk.

For example, BPformer [24] used a Transformer architecture to trace unknown tumor origins. In another study, DeepGene Transformer [25] used a multi-head self-attention mechanism to classify cancer subtypes by identifying biomarkers without prior feature selection. Moreover, GeneViT [26] integrated a Vision Transformer model with the DeepInsight technique to improve cancer classification accuracy by representing gene expression data as images. These studies used Transformer-based models for gene expression analysis but neglected the advantages of multi-task learning.

B. Multi-Task Learning

Multi-task learning (MTL) [27] is a method that simultaneously optimizes a set of related tasks, employing either hard or soft parameter sharing. The primary objective of MTL is to capitalize on the valuable information inherent in these tasks, with the aim of enhancing the generalization performance across all tasks. ML-GCN [28] built a directed graph across different object labels to enhance multi-label image recognition, where each node represents a particular object (i.e., task) and the edges denote object correlations. Dizaji et al. [29] designed a deep multitask learning algorithm to infer gene expression, which leverages a subnetwork with low-dimensional latent variables to capture correlations between target genes, aiming to efficiently predict expressions for approximately target genes. Multi-tissue

transcriptome mapping (MTM) [30] is a deep learning-based multi-task learning framework designed to predict individualized expression profiles across tissues by leveraging individualized cross-tissue information from reference samples, aiming to enhance the accuracy of predictions and preserve individual biological variations. Network-based Semi-supervised Multi-task Learning model (NetSML) [31] identifies common and cancer-specific differentially expressed genes across tumor types, using latent gene co-expression modules and gene-sample biclusters to improve sample classification in multiple cancer datasets. However, most methods fail to fully exploit the distinct and valuable features inherent across various tasks, especially in the domain of disease diagnosis and prognosis.

III. PROPOSED METHODOLOGY

Fig. 1 shows the architecture of BioMTAN which contains four modules: Biological Knowledge-guided Graph Construction, Multi-level Graph Learning Module, Multi-task Attention Module and Multi-task Prediction Module. Given gene expression data X , BioMTAN predicts the labels of two tasks simultaneously: molecular subtype label $\hat{Y}_{\text{subtype}}$ and survival risk label $\hat{Y}_{\text{risk}}$. Algorithm 1 is the corresponding pseudo-code.

Specifically, we construct graph sets G_s and G_r for these two tasks, guided with gene expression data and prior biological knowledge (i.e., biological pathways collections). Through the

Multi-level Graph Learning Module, we extract multi-level gene information from each subgraph and concatenate these feature representations across different levels along the feature dimension to produce biological pathway feature representation sets F_s and F_r , and serve F_s/F_r as task-specific local information. The Multi-task Attention Module, composed of a Global Learning Block and a Task-driven Learning Block, enables interactions among biological pathways and between different tasks. F_s and F_r are fed into the Global Learning Block to derive global key information F_g . The Task-driven Learning Block utilizes task-specific local information as the query and global information as both the key and value, effectively decoding task-awareness features. Finally, the predicted labels $\hat{Y}_{\text{subtype}}$ and $\hat{Y}_{\text{risk}}$ are output from the Multi-task Prediction Module.

Further details of each module and the optimization objective are elaborated below.

A. Biological Knowledge-Guided Graph Construction

Biological pathways can be considered as gene networks consisting of a set of genes as nodes and their interactions as edges [32], [33], collaboratively executing specific biological functions. For various disease-related tasks, the essential genes and biological pathways typically exhibit significant variation [14], [15]. Recognizing this variability, we harness prior knowledge to meticulously construct tailored pathway sets specific to each task, ensuring enhanced relevance and accuracy in subsequent analyses.

To develop a BioMTAN in this study, we select two sets of pathways P_s and P_r , which are highly correlated with cancer subtype classification and survival risk prediction, respectively. Each biological pathway inherently includes a set of genes (nodes) and the interactions (edges) between them. With biological knowledge guidance, we construct two sets of graphs $G_s = \{G_s^1, G_s^2, \dots, G_s^m\}$ and $G_r = \{G_r^1, G_r^2, \dots, G_r^n\}$. G_s is constructed with m biological pathways in P_s highly correlated with cancer classification, and G_r is constructed with n biological pathways in P_r highly correlated with survival risk. For each pathway, a graph G_s^i/G_r^i is created where nodes (vertices) represent genes in the pathway P_s^i/P_r^i , edges represent the interactions or relationships between these genes in P_s^i/P_r^i . In the case of template graph set G_s and G_r , the i -th selected pathway graph can be described as $G_s^i = (V_s^{M \times 1}, E_s^{M \times M})$ and $G_r^i = (V_r^{N \times 1}, E_r^{N \times N})$, where $V_s^{M \times 1}$ represents the features of M nodes and the edge matrix $E_s^{M \times M}$ denotes the edge connections in the i -th pathway in P_s , $V_r^{N \times 1}$ represents the features of N nodes, with each node contains the expression of a gene, and the edge matrix $E_r^{N \times N}$ denotes the edge connections in the i -th pathway in P_r . In the edge matrices $E_s^{M \times M}$ and $E_r^{N \times N}$, an entry is 1 if there is an interaction in KEGG database in between the corresponding pair of nodes (genes) and 0 if there is no interaction.

$$E_{ij} = \begin{cases} 1 & \text{if genes } i \text{ and } j \text{ interact in KEGG,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Algorithm 1: Pseudo Code of Proposed Methodology.

```

1: Input:
2:  $X$ : gene expression data
3:  $P_s$ : pathways set correlated with subtype classification
4:  $P_r$ : pathways set correlated with survival risk prediction
5: Output:
6:  $\hat{Y}_{\text{subtype}}$ : predicted molecular subtype label
7:  $\hat{Y}_{\text{risk}}$ : predicted survival risk label
8: 1. Biological Knowledge-guided Graph Construction
9:  $G_s = []$  # Initialize graph set for classification pathways
10:  $G_r = []$  # Initialize graph set for survival risk pathways
11: for each PathwaySet, GraphSet in  $[(P_s, G_s), (P_r, G_r)]$  do:
12:   for each pathway  $P_i$  in PathwaySet do:
13:     # Genes is the list of genes in the pathway
14:     Genes = ExtractGenesFromPathway( $P_i$ )
15:     #  $V$  is a list of expression values for each gene
16:      $V = [X[g]$  for  $g$  in Genes]
17:      $E = \text{EmptyMatrix}(\text{len}(\text{Genes}), \text{len}(\text{Genes}))$ 
18:     for  $a$  in  $\text{range}(\text{len}(\text{Genes}))$  do:
19:       for  $b$  in  $\text{range}(\text{len}(\text{Genes}))$  do:
20:         if InteractionExists(Genes[ $a$ ], Genes[ $b$ ]) in pathway  $P_i$  then:
21:            $E[a][b] = 1$ 
22:         else:
23:            $E[a][b] = 0$ 
24:         end if
25:       end for
26:     end for
27:     Append  $G_i = (V, E)$  to GraphSet
28:   end for
29: end for
30: 2. Multi-level Graph Learning Module
31:  $F_s = []$  # Features for classification task
32:  $F_r = []$  # Features for survival risk task
33: for each GraphSet, FeatureSet in  $[(G_s, F_s), (G_r, F_r)]$  do:
34:   for each graph  $G_i$  in GraphSet do:
35:     # MFE is Multi-level Feature Extractor
36:     features = MFE( $G_i$ )
37:     Append features to FeatureSet
38:   end for
39: end for
40: 3. Multi-task Attention Module
41:  $F_g = \text{GL}([F_s, F_r])$  # GL is Global Learning Block
42:  $T_s = \text{TL}(F_s, F_g)$  # TL is Task-driven Learning Block
43:  $T_r = \text{TL}(F_r, F_g)$ 
44: 4. Multi-task Prediction Module
45:  $\hat{Y}_{\text{subtype}} = \text{MLP}(T_s, \theta_{\text{Spred}})$ 
46:  $\hat{Y}_{\text{risk}} = \text{MLP}(T_r, \theta_{\text{Rpred}})$ 

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B. Multi-Level Graph Learning Module

As reported in [22], the full fusion of multi-level graph features can produce more comprehensive and discriminative gene features. In addition, the output feature of each node after the Graph Attention Network (GAT) layer is a weighted combination of the gene features on the co-functional gene modules, which can better reflect disease status. Therefore, we follow this paradigm and develop the Multi-level Graph Learning Module to learn each graph in G_s and G_r . This module comprises of Multi-level Feature Extractors (MFE) with distinct parameters to learn individual subgraphs effectively. This is crucial as different biological pathways involve unique genes and interactions between these genes.

Specifically, we have

$$G_2^i = \text{GAT}(G^i, \omega_1^g) \quad (2)$$

$$G_3^i = \text{GAT}(G_2^i, \omega_2^g) \quad (3)$$

$$G_2^{i'} = G_2^i \cdot \theta_{\text{MFE}}^1 \quad (4)$$

$$G_3^{i'} = G_3^i \cdot \theta_{\text{MFE}}^2 \quad (5)$$

$$F^i = \text{MLP}([\text{Vec}(G^i), \text{Vec}(G_2^{i'}), \text{Vec}(G_3^{i'})], \theta_{\text{MFE}}^3) \quad (6)$$

where $G^i = (V^{K \times 1}, E^{K \times K})$ represents the single graph with K nodes in G_s or G_r . $G_2^i = (V^{K \times h_2}, E^{K \times K})$ and $G_3^i = (V^{K \times h_3}, E^{K \times K})$ are high-level graphs generated by G^i with two GAT layers with weight parameters ω_1^g and ω_2^g . Multi-Layer Perceptron (MLP) with parameters $\theta_{\text{MFE}}^1 \in \mathbb{R}^{h_2}$ and $\theta_{\text{MFE}}^2 \in \mathbb{R}^{h_3}$ following by GATs are performed to reduce node dimension of G_2^i and G_3^i , resulting in $G_2^{i'} = (V^{K \times 1}, E^{K \times K})$ and $G_3^{i'} = (V^{K \times 1}, E^{K \times K})$. Each MLP is followed by a ReLU activation function. Then, $G_2^{i'}$ and $G_3^{i'}$ are vectorized and concatenated to yield the fused feature. Finally, we utilize MLP with parameters $\theta_{\text{MFE}}^3 \in \mathbb{R}^{3K \times D}$ to unify the dimension of fused features in each graph in G_s and G_r and obtain the biological feature representation $F^i \in \mathbb{R}^D$. Here, ω_1^g and ω_2^g are learnable weight parameters in the GAT layers. The GAT operation computes node representations by aggregating neighbor features using attention scores. $\text{Vec}(\cdot)$ denotes a flattening operation applied to the node features of a graph.

The Multi-level Graph Learning Module encodes each pathway graph into fine-grained representations, which are then used by the Multi-task Attention Module to effectively capture shared and task-specific representations.

C. Multi-Task Attention Module

To effectively leverage the shared and task-specific patterns in gene expression data, the Multi-task Attention Module is designed to balance information sharing and task distinction. It consists of two complementary components: the Global Learning Block, which extracts shared global information across tasks, and the Task-driven Learning Block, which refines this information for each individual task. Within the Global Learning Block, features from different tasks $F_s \in \mathbb{R}^{m \times D}$ and $F_r \in \mathbb{R}^{n \times D}$ (i.e., distinct pathway sets) are fused and extensively learned to form global information F_g . The Task-driven Learning Block, taking

the pivotal local information F_s/F_r as query while global information F_g as key and value, can enrich the prediction process for specific tasks with more relevant and effective information T_s/T_r .

Both the Global Learning Block and the Task-driven Learning Block are composed of two parts, i.e., a multiheaded self-attention (MSA) and an MLP block. Layer normalization (LN) was applied before each block, and residual connections were incorporated after each block.

The input feature of MSA is projected onto the query (Q), key (K) and value (V) of dimension d_k and self-attention is being computed by the Q , K and V . For a pathway set of length l , the projections yield $Q \in \mathbb{R}^{l \times d_k}$, $K \in \mathbb{R}^{l \times d_k}$ and $V \in \mathbb{R}^{l \times d_v}$. The MSA with k heads extended self-attention (SA) by concurrently executing k self-attention operations, and then projected their concatenated results. The MSA operator is calculated as:

$$\text{SA}(Q, K, V) = V \cdot \text{Softmax}\left(\frac{Q \cdot K^T}{d_k}\right) \quad (7)$$

$$\text{MSA}(Q, K, V) = [\text{SA}_1(Q, K, V); \dots; \text{SA}_k(Q, K, V)] \quad (8)$$

Note that the query features of the Global Learning Block and the Task-driven Learning Block are fundamentally different. Unlike previous models that primarily rely on general representations, our approach integrates task-specific pathway knowledge into a Transformer-based architecture. The query features in the Global Learning Block and Task-driven Learning Block are designed to incorporate biologically relevant information, providing task-specific guidance. This explicit integration of pathway-level knowledge not only enhances the model's interpretability but also improves its performance.

1) Global Learning Block: We first concatenate the features F_s and F_r as $F_{sr} \in \mathbb{R}^{(m+n) \times D}$ from the Multi-level Graph Learning Module output and project them onto the query (Q), key (K) and value (V) with weight parameters $W^Q, W^K, W^V \in \mathbb{R}^{D \times \frac{D}{(m+n)}}$ in MSA. Specifically, we have:

$$F_{sr} = \text{Concat}(F_s, F_r) = [F_s, F_r] \quad (9)$$

$$Q, K, V = \text{LN}(F_{sr} \cdot W^Q), \text{LN}(F_{sr} \cdot W^K), \text{LN}(F_{sr} \cdot W^V) \quad (10)$$

$$F'_{sr} = \text{MSA}(Q, K, V) + F_{sr} \quad (11)$$

$$F_g = \text{MLP}(\text{LN}(F'_{sr}, \theta_{\text{GL}})) + F'_{sr} \quad (12)$$

where θ_{GL} is the weight parameters of MLP in Global Learning Block. This block outputs the global information F_g .

2) Task-Driven Learning Block: Unlike the Global Learning Block which uses the same feature for the query, key, and value, we project the task-specific local information F_s/F_r with weight parameters $W^{\hat{Q}}$ as the query $\hat{Q}$, while projecting the global information F_g with weight parameters $W^{\hat{K}}$ and $W^{\hat{V}}$ as the key $\hat{K}$ and value $\hat{V}$ in the MSA of the Task-driven Learning Block to decode the task-aware features $T_{s/r}$. Specifically, we

have:

$$\hat{Q}, \hat{K}, \hat{V} = \text{LN}(F_{s/r} \cdot W^{\hat{Q}}), \text{LN}(F_g \cdot W^{\hat{K}}), \text{LN}(F_g \cdot W^{\hat{V}}) \quad (13)$$

$$F'_{s/r} = \text{MSA}(\hat{Q}, \hat{K}, \hat{V}) + F_{s/r} \quad (14)$$

$$T_{s/r} = \text{MLP}(\text{LN}(F'_{s/r}, \theta_{\text{TL}})) + F'_{s/r} \quad (15)$$

where $W^{\hat{Q}}, W^{\hat{K}}, W^{\hat{V}} \in \mathbb{R}^{D \times \frac{D}{(m+n)}}$ in MSA and θ_{TL} is the weight parameters of MLP in Task-driven Learning Block.

D. Multi-Task Prediction Module

After the Multi-task Attention Module, the task-specific representation T_s/T_r are fed into standard classifiers include MLP heads followed with the ReLU activation function for each branch to predict the task labels.

The label predictions $\hat{Y}_{\text{subtype}}$ and $\hat{Y}_{\text{risk}}$ can be denoted as

$$\hat{Y}_{\text{subtype}} = \text{MLP}(T_s, \theta_{\text{Spred}}) \quad (16)$$

$$\hat{Y}_{\text{risk}} = \text{MLP}(T_r, \theta_{\text{Rpred}}) \quad (17)$$

where θ_{Spred} and θ_{Rpred} are the weight parameters of MLP for predicting cancer subtype and survival risk, respectively.

To train the network, we employ the cross-entropy loss as the loss function of the subtype prediction part $\mathcal{L}_{\text{subtype}}$, and leverage Cox partial likelihood loss [34] with l_1 regularization as the loss function of the risk prediction part $\mathcal{L}_{\text{risk}}$ following [35]. These loss functions align well with the respective tasks: cross-entropy for multi-class classification, and Cox loss for survival prediction, as it effectively handles censored data and models relative risk without requiring a predefined survival time distribution. The total loss $\mathcal{L}$ is defined as the summation of the two loss.

$$\mathcal{L} = \mathcal{L}_{\text{subtype}} + \mathcal{L}_{\text{risk}} \quad (18)$$

IV. EXPERIMENTS AND RESULTS

A. Dataset

TCGA Datasets: We assessed our proposed model using public datasets from The Cancer Genome Atlas (TCGA),¹ with a specific focus on bladder urothelial carcinoma (BLCA), lower grade glioma (LGG), and uterine corpus endometrial carcinoma (UCEC). For each dataset, we meticulously selected subjects having complete data on cancer molecular subtypes, follow-up duration (in months), and status. Concretely, BLCA encompasses five subtypes [36]: Luminal-Papillary (LumP), Luminal-Infiltrated (LumI), Luminal, Neuronal (Neu), and Basal-Squamous (Ba-Sq). LGG includes three subtypes [37]: IDH wildtype (IDHwt), IDH mutant with 1p/19q codeletion (IDHmut-codel), and IDH mutant without 1p/19q codeletion (IDHmut-non-codel). The UCEC dataset comprises four subtypes [38]: POLE ultramutated (POLE), Microsatellite instability hypermutated (MSI), Copy-number high, serous-like (CN-High) and Copy-number low, endometrioid (CN-Low). A detailed summary of these datasets is provided in Table I.

TABLE I

THE BASIC CHARACTERISTICS OF DATASETS, INCLUDING PATIENT COUNT, SUBTYPE CATEGORIES AND QUANTITIES, FOLLOW-UP TIME (MONTH), STATUS (ALIVE/DEAD)

Dataset	Patient	Subtype	Follow-up	Status
BLCA	231	Ba-Sq, LumP, LumI, Luminal, Neu (82/81/44/15/9)	22.10±22.61	134/97
LGG	503	IDHwt, IDHmut-codel, IDHmut-non-codel (95/165/243)	31.85±31.77	380/123
UCEC	502	MSI, POLE, CN-LOW, CN-HIGH (147/49/144/162)	38.14±30.15	422/80

Incorporation of Biological Pathways: Prior biological knowledge (i.e., biological pathway collections) are derived from the Kyoto Encyclopedia of Genes and Genomes database (KEGG,² accessed 5 February 2023). In our study on cancer molecular subtype classification, we selected 11 pathways categorized as “cancer overview” from the KEGG database, along with one pathway specifically expressed in certain cancer types. These 11 pathways are involved in key mechanisms of cancer initiation, progression, and metastasis, comprehensively cover the common molecular mechanisms of different cancer types, and have broad significance in different cancers [10]. Additionally, we included one pathway specifically associated with a certain type of cancer. In the KEGG database, 45 pathways are annotated as “signaling pathways”, including several key oncogenic signaling pathways such as Wntless/Int-1 (WNT), Hippo, Notch, Phosphatidylinositol 3-Kinase - Protein Kinase B (PI3K-Akt), Rat Sarcoma (RAS), Transforming Growth Factor-beta (TGF-beta), and Tumor Protein p53 (p53) [15]. In addition, the “cell cycle” cellular process is also included and simplified as a “signaling pathway” for convenience. Therefore, a total of 46 pathways (45 signaling pathways plus cell cycle) are selected to constitute the survival-related pathway set [14]. Consequently, a total of 46 signaling pathways (comprising the 45 signaling pathways and the cell cycle) are selected to constitute the Survival-related pathway set. As a result, the first prior knowledge set for cancer subtype prediction contains 12 pathways and the second prior knowledge set for survival risk prediction includes 46 pathways.

B. Implementation Details

We developed BioMTAN using PyTorch v2.0.1 in Python v3.8.13. During the training phase, the Adam optimizer was employed for parameter optimization, with a batch size of 32, across 50 epochs. We implemented a cosine annealing strategy, starting with an initial learning rate of 0.001, progressively reduced to a minimum of 0.0001. To ensure robustness, we applied 5-fold cross-validation for training all models.

¹<https://portal.gdc.cancer.gov/>

²<https://www.genome.jp/kegg/>

TABLE II
MODEL PERFORMANCE ON SUBTYPE AND RISK PREDICTION

Dataset	Subtype prediction			Risk prediction		
	Acc	Precision	F1	C-Index	p-value	FDR
BLCA	0.9015±0.0342	0.8808±0.0416	0.8821±0.0377	0.6359±0.0390	3.674e-08	3.6740e-08
LGG	0.9663±0.0311	0.9675±0.0296	0.9665±0.0307	0.8421±0.0271	4.441e-16	1.3323e-15
UCEC	0.8172±0.0417	0.8251±0.0388	0.8119±0.0434	0.7038±0.0350	1.609e-10	2.4135e-10

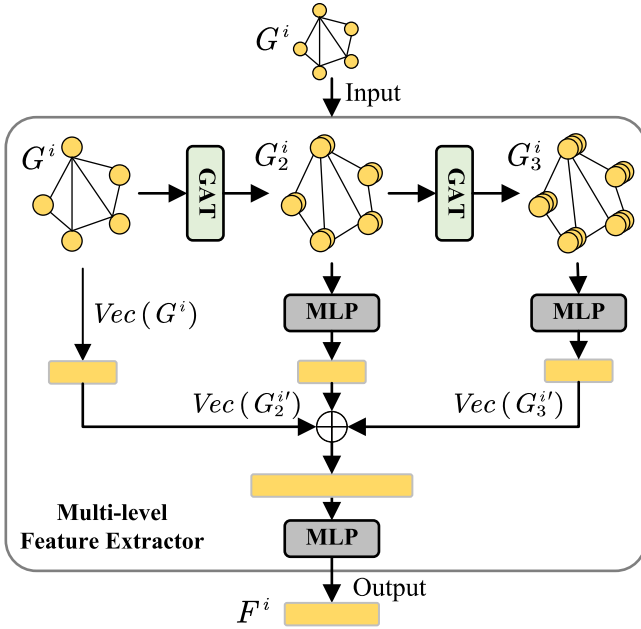


Fig. 2. The structure of Multi-level Feature Extractor (MFE) in Multi-level Graph Learning Module.

C. Evaluation Metrics

For the cancer subtype prediction task, we employed a range of metrics to ensure a comprehensive evaluation. Specifically, we utilized Accuracy (Acc), Precision and F1-score (F1) as the primary measures. The survival prediction task was assessed using the C-Index, a metric that gauges the alignment between the forecasted risk and actual survival outcomes. For statistical analysis, we also calculated a two-sided p value, considering values less than 0.05 as indicative of a statistically significant difference, and used the Benjamini-Hochberg procedure to adjust the false discovery rate (FDR).

D. Multi-Task Performance

1) **Subtype Prediction Performance:** Table II shows the cancer subtype prediction performance of our study. Our proposed model achieved an average accuracy of 0.9015 for BLCA, which was consistent with a precision of 0.8808 and an F1 score of 0.8821. In the case of LGG, the model demonstrated a higher accuracy of 0.9663, with a precision of 0.9675 and an F1 score of 0.9665. For the UCEC dataset, the model attained an accuracy of 0.8172, with corresponding precision and F1 scores of 0.8251

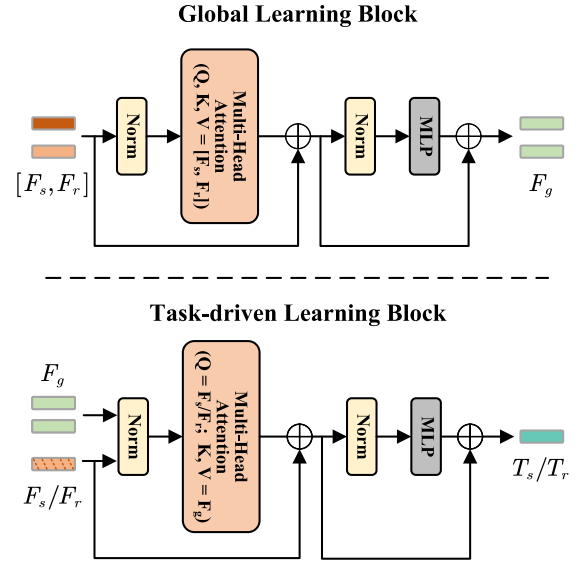


Fig. 3. The structure of the Multi-task Attention Module, including the Global Learning Block and the Task-driven Learning Block.

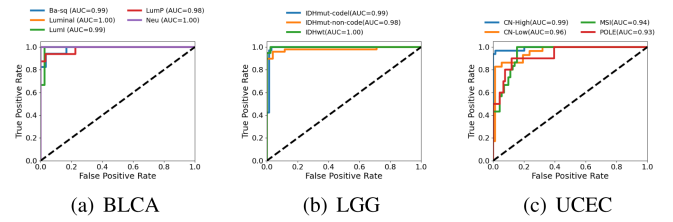


Fig. 4. AUC-ROC plots for subtype prediction across different cancer types.

and 0.8119, respectively. These results suggest that our model not only distinguishes between cancer molecular subtypes with high accuracy but also maintains a robust balance between precision and F1 scores. To provide an overall insight into the model's performance across both classes, we generated AUC-ROC plots for subtype prediction across different cancer types, as shown in Fig. 4. We selected the best-performing model and its corresponding data from the 5-fold cross-validation experiments to create these plots. For BLCA, the model achieved high AUC values for all subtypes, with perfect scores ($AUC = 1.00$) for Neu and Luminal subtypes. Similarly, for LGG, the model demonstrated strong predictive performance with AUC values near or at 1.00, effectively distinguishing between IDHmut-code, IDHmut-non-code, and IDHwt subtypes. In the case of

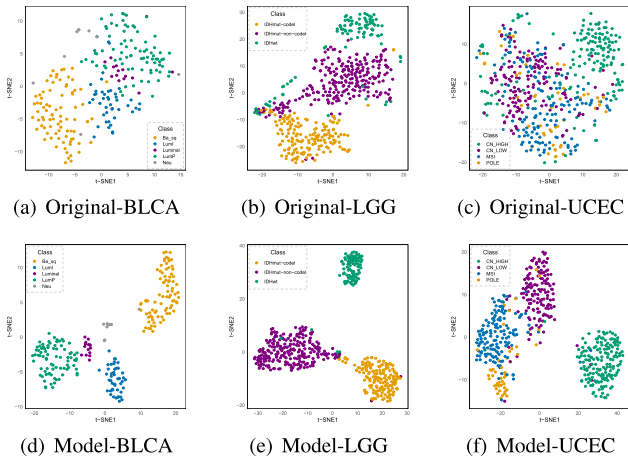


Fig. 5. The t-SNE visualizations of cancer subtype classifications for different datasets. (a) to (c) display the original sample distributions for BLCA, LGG, and UCEC, respectively. (d) to (f) illustrate the sample distributions as computed by our model for the corresponding cancer types.

UCEC, the model also demonstrated effective performance, with slightly lower AUC values around 0.9 for CN-Low and POLE subtypes. Overall, these results highlight the model's robustness in predicting various cancer subtypes and its potential utility in diagnostic and oncological research.

To more intuitively assess the validity of the representations learned by BioMTAN, we employed t-SNE [39] for visualization. The representations derived from the BioMTAN model, as depicted in Fig. 5(d) and (e), demonstrate more distinct and well-defined clusters for BLCA and LGG when compared to the original sample representations. Fig. 5(f) also presents distinct clusters, although there are some overlaps. Overall, the t-SNE visualizations underscore BioMTAN's efficacy in enhancing the clarity and distinction of cancer subtype predictions when compared to the original data representations.

2) Survival Risk Prediction Performance: The results presented in Table II highlight the model's statistically significant predictive prowess across different datasets. For the BLCA dataset, a C-Index of 0.6359 indicates the model's reliable prognostic ability. This is further evidenced by the LGG dataset's C-Index of 0.8421, reflecting a superior predictive performance. The UCEC dataset exhibits a C-Index of 0.7038. Together, these metrics affirm the model's consistent precision in predicting survival risks associated with different cancer subtypes.

To further validate the effectiveness of BioMTAN for survival analysis, we stratified all patients into low risk group (yellow) and high risk group (blue) according to the mid-value of the predicted risk scores from BioMTAN. After that, we utilized Kaplan-Meier analysis with Logrank test (p-value) to visualize the survival events of all patients, analysis results are shown in Fig. 6. For the BLCA dataset, the survival probability of the low-risk group remained notably higher than that of the high-risk group over the follow-up period, with a significant p-value of $3.674e-08$. Similarly, in the LGG dataset, the low-risk group exhibited a pronouncedly higher survival

probability, with the difference validated by a compelling p-value of $4.441e-16$. For the UCEC dataset, the difference remained statistically significant with a p-value of $1.609e-10$. We calculated the FDR using the Benjamini-Hochberg procedure to evaluate significant risk prediction across all datasets for multiple testing, resulting in FDR-adjusted p-values of $3.6740e-08$ for BLCA, $1.3323e-15$ for LGG, and $2.4135e-10$ for UCEC (TABLE II). These survival curves emphasize the model's capacity to effectively stratify patients based on risk and predict reliable survival outcomes across various datasets.

E. Model Comparison

Previous methods overlooked the use of prior specific pathway collections for subtype prediction and survival risk prediction. Therefore, this study utilizes mainstream CNNs such as VGG-19 [40], ResNet-50 [41], DenseNet-121 [42], Inception-v4 [43] and Xception [44] along with Transformer architectures [45] and advanced graph neural networks, including the graph convolutional network (GCN) [46] and Graph attention networks (GAT) [47]. Additionally, models specifically designed for gene expression data, such as 1D-CNN [48] and Deep-Gene Transformer [25], were also tested. These technologies form the foundation of our baselines, which we categorized into two distinct groups: non-graph-based and graph-learning-based methods. For model comparison, we present accuracy (Acc), true positive rate (TPR), and false positive rate (FPR) for subtype prediction, along with the C-index for risk prediction.

In graph-learning-based methods, subgraphs were derived from biological pathways and analyzed using graph neural networks (GNNs) for feature extraction. Specifically, the GNN processes each pathway as a distinct subgraph, subsequently aggregating the features from this collection of subgraphs for prediction through an MLP. In contrast, non-graph-based methods assessed the model's performance by integrating prior pathway knowledge and comparing it to the primary whole gene expression data. In scenarios lacking prior knowledge, all gene expression data were directly input into the baseline model for prediction. When incorporating two separate sets of prior knowledge, independent CNNs were employed to extract features from the gene expression data of each pathway. Alternatively, these pathways were processed as tokens in a Transformer model, with features from each set subsequently merged for the final prediction.

Table III delineates a thorough performance comparison. For Non-graph-based methods, we find that using genes from prior knowledge greatly improves model performance. Specially, the Transformer model notably improved subtype prediction accuracy by 11.17%, 17.71%, and 11.44% on the BLCA, LGG, and UCEC datasets, respectively. It also enhanced the C-Index by 6.18%, 10.77%, and 5.68% for these datasets. Furthermore, the use of GNNs or Transformer architecture, informed by the collection of prior knowledge, shows superior performance compared to CNNs. This enhanced performance is likely due to the ability of GNNs to more effectively capture gene relationships, and Transformers' focus on pathway interactions. Our proposed model incorporates these elements and integrates

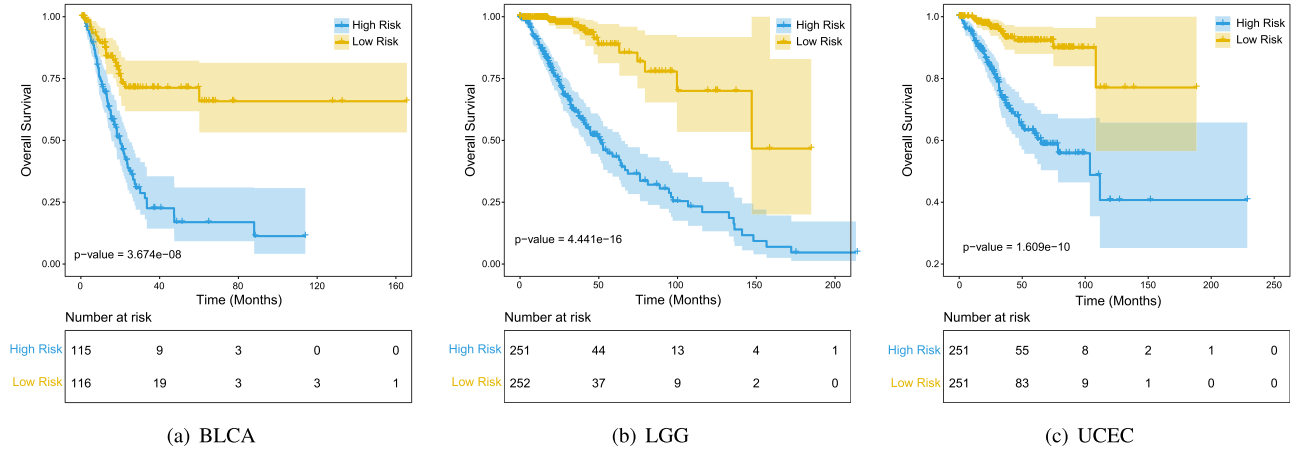


Fig. 6. According to predicted risk scores, all patients are stratified into low risk group (yellow) and high risk group (blue). Kaplan-Meier analysis and Logrank test (p-value) are utilized to measure the statistical significance between low risk group and high risk group.

TABLE III
THE PERFORMANCE COMPARISON BETWEEN BASELINE MODELS AND PROPOSED MODEL ON SUBTYPE AND SURVIVAL PREDICTION

	Method	BLCA				LGG				UCEC			
		Acc	TPR	FPR	C-Index	Acc	TPR	FPR	C-Index	Acc	TPR	FPR	C-Index
Non-graph-based	VGG (-pn) ¹	54.80±7.75	38.65±8.69	19.65±5.64	0.5661±0.0505	64.35±8.06	56.96±8.99	18.47±6.78	0.6501±0.0630	51.67±3.12	38.44±3.21	28.64±2.23	0.5321±0.0314
	VGG (+pn) ²	69.39±8.74	47.45±12.05	9.35±3.19	0.6162±0.0413	78.51±11.63	65.56±10.54	15.49±9.58	0.7755±0.0289	60.69±5.04	49.02±1.06	24.76±0.48	0.6189±0.0152
	ResNet (-pn) ¹	60.23±7.14	41.63±5.41	15.96±6.23	0.5521±0.3760	70.61±4.13	57.69±5.69	15.66±6.35	0.6511±0.0761	51.02±3.32	40.69±2.69	20.98±2.39	0.5215±0.0318
	ResNet (+pn) ²	67.39±3.02	44.96±8.64	12.36±5.23	0.6271±0.0287	83.21±8.72	71.85±4.69	8.66±1.01	0.7779±0.0380	66.61±2.32	57.01±3.25	11.63±1.44	0.6188±0.0431
	DenseNet (-pn) ¹	60.26±4.56	43.29±3.69	14.11±4.36	0.5703±0.0481	68.41±3.64	63.45±4.36	14.96±4.65	0.6302±0.0964	53.61±6.11	36.97±2.14	19.67±1.66	0.5421±0.0351
	DenseNet (+pn) ²	75.66±6.82	67.33±5.21	6.99±3.63	0.6241±0.0515	76.98±6.27	65.99±4.23	9.68±2.68	0.7019±0.0350	70.97±2.63	62.87±4.98	10.11±1.58	0.6619±0.0532
	Inception (-pn) ¹	62.32±6.32	45.96±6.33	10.66±2.78	0.5652±0.0413	68.42±3.62	62.39±1.69	14.44±3.69	0.6220±0.0260	55.32±3.65	42.63±2.39	17.85±1.67	0.5244±0.0332
	Inception (+pn) ²	69.25±3.96	47.59±9.23	10.48±3.06	0.5961±0.0197	74.46±3.93	66.28±3.56	10.32±1.58	0.7220±0.0585	63.54±5.62	54.48±7.21	16.14±2.02	0.6006±0.0434
	Xception (-pn) ¹	69.43±4.23	55.43±2.96	6.42±2.31	0.5644±0.0460	70.91±6.27	63.25±2.36	11.66±2.51	0.6621±0.0512	58.12±4.36	48.96±4.14	14.59±1.09	0.5409±0.0356
	Xception (+pn) ²	79.12±6.47	72.17±4.22	4.23±0.29	0.6033±0.0305	76.69±5.45	67.95±2.96	8.69±0.96	0.7903±0.0250	65.60±8.32	57.63±3.26	13.66±1.96	0.6291±0.0402
	Transformer (-pn) ¹	70.01±6.23	60.94±5.36	6.84±2.01	0.5612±0.0641	75.12±4.23	75.03±0.96	8.45±1.23	0.7065±0.0502	67.51±3.21	63.98±6.35	7.96±5.22	0.6211±0.0401
	Transformer (+pn) ²	81.18±8.21	72.69±6.99	3.68±1.56*	0.6230±0.0602	92.83±2.92*	90.03±0.81	2.43±0.37	0.8142±0.0453	78.95±5.66*	71.56±2.02*	7.15±1.56*	0.6779±0.0444
	ID-CNN	78.62±9.40	72.78±5.63	5.96±2.63	0.5922±0.0550	80.62±9.39	77.42±1.96	8.96±1.69	0.7427±0.0703	69.33±5.87	67.06±4.36	7.98±3.21	0.6559±0.0542
	DeepGene Transformer	80.51±4.65	68.77±4.69	5.66±2.41	0.5761±0.0408	80.86±2.55	78.64±1.45	8.69±1.21	0.7462±0.0325	73.69±3.65	70.64±3.25	8.41±3.42	0.6459±0.0364
Graph-learning-based	GCN	83.64±6.97	73.39±8.10*	3.77±1.48	0.6315±0.0190	91.68±4.27	89.81±1.07	2.47±0.54	0.8209±0.0528	73.36±3.44	66.07±4.21	7.49±1.48	0.6826±0.0382
	GAT	84.02±3.24*	72.19±8.45	3.80±1.13	0.6333±0.0458*	91.46±3.04	90.72±1.67*	2.03±0.75*	0.8262±0.0517*	71.60±4.53	65.67±5.96	7.18±1.29	0.6837±0.0579*
	BioMTAN	90.15±3.42	81.25±3.63	2.24±0.71	0.6359±0.0390	96.63±3.11	95.11±1.72	1.68±0.73	0.8421±0.0271	81.72±4.17	73.52±5.88	6.49±2.04	0.7038±0.0350

¹ (-pn) means train and validate the model with whole gene expression data.

² (+pn) means train and validate the model with gene expression data with prior knowledge (pn) of biological pathway collections.

* refers to the second-best results and the best results are highlighted in bold.

Multi-task attention, resulting in the most robust performance and significantly exceeding the second-best results. It achieved an improvement of 6.13% in accuracy, 7.86% in TPR, 1.44% in FPR, and 0.26% in C-Index on the BLCA dataset; a gain of 3.80% in accuracy, 4.39% in TPR, 0.35% in FPR, and 1.59% in C-Index on the LGG dataset; and an enhancement of 2.77% in accuracy, 1.96% in TPR, 0.66% in FPR, and 2.01% in C-Index on the UCEC dataset.

F. Ablation Study

In order to investigate the effectiveness of different parts of our model, we conducted ablation study on the UCEC dataset. Table IV presents the results of our ablation study, meticulously dissecting the influence of three components: Multi-Graph Learning (MGL) Module, Global Learning (GL) Block, and Task-driven Learning (TL) Block. Method M1, leveraging only the MGL component, achieved an accuracy of 0.7657 and a C-Index of 0.6849. When combining MGL with GL, as in M2,

TABLE IV
THE ABLATION EXPERIMENT ON UCEC. MGL, GL AND TL CORRESPOND TO MULTI-GRAPH LEARNING MODULE, GLOBAL LEARNING.

Method	Block, and Task-driven Learning Block			Acc	C-Index
	MGL	GL	TL		
M1	✓			0.7657±0.0790	0.6849±0.0361
M2	✓	✓		0.7718±0.0935	0.6976±0.0377
M3	✓		✓	0.7895±0.0596	0.6900±0.0357
M4	✓	✓	✓	0.8172±0.0417	0.7038±0.0350

there was a marked improvement in both metrics, indicating an accuracy of 0.7718 and a C-Index of 0.6976. This underscores the synergistic effect of employing Global Learning strategy, showing the effectiveness of introducing more abundant information within cancer diagnosis and prognosis. In contrast, the incorporation of TL alongside MGL in method M3 yielded an accuracy of 0.7895 and a C-Index of 0.6900, indicating that while

TABLE V
COMPARISON OF SINGLE-TASK AND MULTI-TASK PARADIGM

Data	Paradigm	Acc	C-Index
BLCA	Single	0.8892±0.0361	0.6213±0.0425
	Multi	0.9015±0.0342	0.6359±0.0390
LGG	Single	0.9384±0.0340	0.8318±0.0292
	Multi	0.9663±0.0311	0.8421±0.0271
UCEC	Single	0.7850±0.0440	0.6781±0.0492
	Multi	0.8172±0.0417	0.7038±0.0350

Task-driven Learning brings value to the model, its integration with MGL without GL doesn't surpass the performance of the M2 configuration. Notably, the most comprehensive method M4, which amalgamates all three components (MGL, GL, and TL) outperforms other configurations, registering an accuracy of 0.8172 and a C-Index of 0.7038. This signifies the cumulative benefits of leveraging all three techniques in tandem, because of the introduction of richer global feature information simultaneously suppressing some irrelevant information (i.e., noise). The ablation analysis clearly elucidates the individual and combined contributions of MGL, GL, and TL, affirming the necessity of their integration for optimized performance in the context of the UCEC dataset.

G. Investigation of Multi-Task Learning Paradigm

The Table V offers an in-depth analysis contrasting the performance outcomes between single-task and multi-task learning paradigms across three datasets. In single-task learning, subtype prediction and risk prediction are conducted independently. For the BLCA dataset, the single-task paradigm yielded an accuracy of 0.8892 and a C-Index of 0.6213. However, when transitioning to the multi-task paradigm, there was a noticeable enhancement in both metrics, with an increase of 1.23% in accuracy and 1.46% in C-Index. A similar trend is observed in the LGG dataset. The single-task approach achieved an accuracy of 0.9384 and a C-Index of 0.8318, while the multi-task paradigm showed improvements of 2.79% in accuracy and 1.03% in C-Index. For the UCEC dataset, the single-task paradigm reported an accuracy of 0.7850 and a C-Index of 0.6781, whereas the multi-task approach demonstrated a significant enhancement with increases of 3.22% in accuracy and 2.57% in C-Index. These findings unequivocally indicate that the multi-task learning paradigm consistently outperforms its single-task counterpart across all datasets, highlighting the inherent benefits and potential of employing multi-task learning strategies in cancer diagnosis and prognosis.

H. Interpretability

Diagnosis: To elucidate BioMTAN's decision making, we used attention maps to highlight the significant pathways in different tasks. For subtype classification, attention weights revealed distinct pathway contributions in BLCA, LGG and UCEC (Fig. 7). In the BLCA, the pathway hsa05219 exhibited

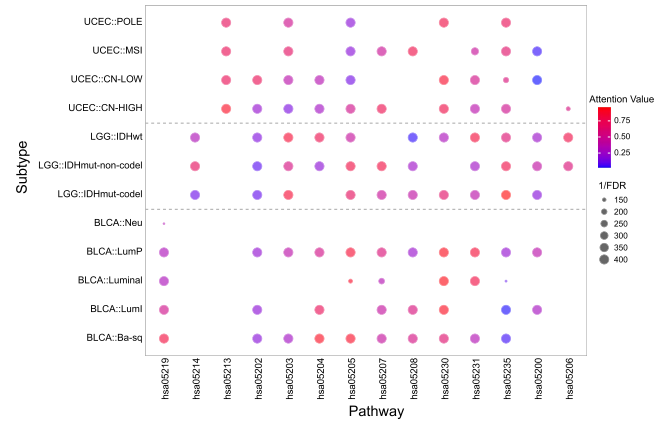


Fig. 7. The interperability of BioMTAN in cancer subtype classification. The dot plot illustrates the importance weights of 14 cancer-related pathways across various cancer subtypes. Colors in the plot represent the mean attention values of each pathway within each cancer subtype, with red indicating higher mean attention and purple lower mean attention. The size of the dots signifies the extent of significant attention differences for a subtype compared to others within each cancer.

significant differences in attention weights across five distinct subtypes (T-test, FDR < 0.01, Fig. 7). Notably, hsa05219 demonstrated the highest weight in the Ba-sq subtype, with the most pronounced differences when compared to other subtypes. In the Neu subtype, the weights across various cancer-associated pathways did not show significant differences compared to other cancer subtypes. This may be attributed to the smaller sample size of the Neu subtype. Furthermore, aside from the Neu subtype, all pathways exhibit significant differences in weight across the various subtypes, with particular disparities observed in the contributions of distinct pathways within the LumP and Ba-sq subtypes. For LGG, 12 pathways showed differential effects. The glioma-specific hsa05214 was most prominent in IDHmut-non-codel and least prominent in IDHmut-codel (T-test, FDR < 0.01), with other cancer pathways also varying significantly across LGG subtypes. This indicates a potential link between pathway activity and molecular characteristics of glioma subtypes. In the UCEC subclassification, hsa05213, which is endemic to endometrial cancer, had the greatest attention in CN-HIGH among four subtypes (T-test, FDR < 0.01), underscoring its potential as a subtype-discriminative biomarker. Additionally, the pathways hsa05208 and hsa05206 are significantly overexpressed in the MSI and CN-HIGH subtypes, respectively, compared to other cancer types. These findings underscore BioMTAN's effectiveness in identifying key pathways for targeted therapies, highlighting its role in developing precise treatments and prompting further research into the mechanisms of cancer subtypes.

Prognosis: In the context of prognosis, BioMTAN has identified key pathways for each cancer type. Pathways hsa04151, hsa04915, hsa04722, hsa04020, and hsa04910 were highlighted for BLCA in Fig. 8(a), patients were well clustered in the heatmap of attention weights. For LGG, the BioMTAN identified pathways hsa04371, hsa04622, hsa04919, hsa04390, and

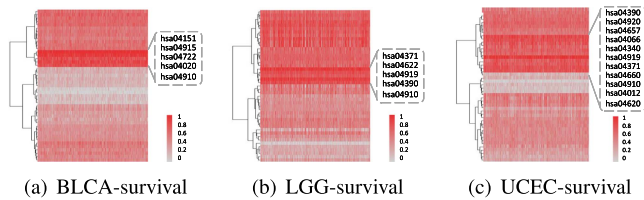


Fig. 8. Heatmap of the attention weight of BioMTAN on cancer prognosis. Pathway attention weights of each patient were extracted from the BioMTAN. Pathways were then reorganized by similarity measure using the ward D.2 clustering.

hsa04910 in Fig. 8(b), which could be integral to patient stratification and treatment planning. For UCEC, the pathways highlighted were hsa04657, hsa04620, hsa04920, hsa04390, and hsa04919 in Fig. 8(c), underscoring their potential role in the pathogenesis and progression of endometrial cancer.

These findings highlight BioMTAN's ability to identify key pathways in cancer diagnosis and prognosis, offering insights for targeted therapies and personalized medicine.

V. CONCLUSION

In this paper, we propose a novel **Biological Knowledge-guided Multi-task Attention Network (BioMTAN)** for cancer diagnosis and prognosis, which capitalizes on the interrelationship between these two areas to promote mutual enhancement. By constructing tailored knowledge collections, we introduce effective gene information and rich biological insights to achieve precise prediction. Furthermore, we develop a Multi-task Attention Learning Module, which enables interactions both among biological pathways and between different tasks. This module adeptly extracts crucial global information from the interactions occurring within biological pathways. Utilizing task-specific local information as the query and global information as both the key and value, it adeptly decodes task-aware features to make prediction for each task. Extensive validation on the public TCGA datasets confirms the superior performance of BioMTAN. Finally, we visualize the attention map to highlight BioMTAN's ability to identify key pathways in disease diagnosis and prognosis. With the increasing availability of high-throughput resources, BioMTAN can potentially be a valuable addition to precision medicine in the clinical setting.

REFERENCES

- [1] D. K. Tobias et al., "Second international consensus report on gaps and opportunities for the clinical translation of precision diabetes medicine," *Nature Med.*, vol. 29, no. 10, pp. 2438–2457, 2023.
- [2] K. A. Tran, O. Kondrashova, A. Bradley, E. D. Williams, J. V. Pearson, and N. Waddell, "Deep learning in cancer diagnosis, prognosis and treatment selection," *Genome Med.*, vol. 13, no. 1, pp. 1–17, 2021.
- [3] A. M. Tsimberidou, E. Fountzilas, L. Bleris, and R. Kurzrock, "Transcriptomics and solid tumors: The next frontier in precision cancer medicine," *Seminars Cancer Biol.*, vol. 84, pp. 50–59, 2022.
- [4] M. Tong et al., "Prioritizing prognostic-associated subpopulations and individualized recurrence risk signatures from single-cell transcriptomes of colorectal cancer," *Brief. Bioinf.*, vol. 24, no. 3, 2023, Art. no. bbad078.
- [5] S. Jubair, A. Alkhateeb, A. A. Tabl, L. Rueda, and A. Ngom, "A novel approach to identify subtype-specific network biomarkers of breast cancer survivability," *Netw. Model. Anal. Health Informat. Bioinf.*, vol. 9, pp. 1–12, 2020.
- [6] L. Zhou, M. Rueda, and A. Alkhateeb, "Classification of breast cancer Nottingham prognostic index using high-dimensional embedding and residual neural network," *Cancers*, vol. 14, no. 4, 2022, Art. no. 934.
- [7] S. Osama, H. Shaban, and A. A. Ali, "Gene reduction and machine learning algorithms for cancer classification based on microarray gene expression data: A comprehensive review," *Expert Syst. With Appl.*, vol. 213, 2023, Art. no. 118946.
- [8] B. M. Kuenzi and T. Ideker, "A census of pathway maps in cancer systems biology," *Nature Rev. Cancer*, vol. 20, no. 4, pp. 233–246, 2020.
- [9] R. Stoney, D. L. Robertson, G. Nenadic, and J.-M. Schwartz, "Mapping biological process relationships and disease perturbations within a pathway network," *NPJ Syst. Biol. Appl.*, vol. 4, no. 1, 2018, Art. no. 22.
- [10] S. Lee, S. Lim, T. Lee, I. Sung, and S. Kim, "Cancer subtype classification and modeling by pathway attention and propagation," *Bioinformatics*, vol. 36, no. 12, pp. 3818–3824, 2020.
- [11] Y. Luo, "SHINE: SubHypergraph inductive neural network," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 18779–18792.
- [12] Y. Wang, Z. Zhang, H. Chai, and Y. Yang, "Multi-omics cancer prognosis analysis based on graph convolution network," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, 2021, pp. 1564–1568.
- [13] T. Wei, S. Wang, J. Zhong, D. Liu, and J. Zhang, "A review on evolutionary multitask optimization: Trends and challenges," *IEEE Trans. Evol. Comput.*, vol. 26, no. 5, pp. 941–960, Oct. 2022.
- [14] J. Feng, H. Zhang, and F. Li, "Investigating the relevance of major signaling pathways in cancer survival using a biologically meaningful deep learning model," *BMC Bioinf.*, vol. 22, no. 1, pp. 1–13, 2021.
- [15] F. Sanchez-Vega et al., "Oncogenic signaling pathways in the cancer genome atlas," *Cell*, vol. 173, no. 2, pp. 321–337, 2018.
- [16] K. R. Kavitha, A. V. Ram, S. Anandu, S. Karthik, S. Kailas, and N. M. Arjun, "PCA-based gene selection for cancer classification," in *Proc. IEEE Int. Conf. Comput. Intell. Comput. Res.*, 2018, pp. 1–4.
- [17] C. Qu et al., "Improving feature selection performance for classification of gene expression data using harris hawks optimizer with variable neighborhood learning," *Brief. Bioinf.*, vol. 22, no. 5, 2021, Art. no. bbab097.
- [18] J. Shen et al., "Deep learning approach for cancer subtype classification using high-dimensional gene expression data," *BMC Bioinf.*, vol. 23, no. 1, pp. 1–17, 2022.
- [19] Y. Zhao et al., "CUP-AI-Dx: A tool for inferring cancer tissue of origin and molecular subtype using RNA gene-expression data and artificial intelligence," *EBioMedicine*, vol. 61, 2020, Art. no. 103030.
- [20] M. Tubishat et al., "Dynamic Salp swarm algorithm for feature selection," *Expert Syst. With Appl.*, vol. 164, 2021, Art. no. 113873.
- [21] L. Meenachi and S. Ramakrishnan, "Metaheuristic search based feature selection methods for classification of cancer," *Pattern Recognit.*, vol. 119, 2021, Art. no. 108079.
- [22] X. Xing et al., "Multi-level attention graph neural network based on co-expression gene modules for disease diagnosis and prognosis," *Bioinformatics*, vol. 38, no. 8, pp. 2178–2186, 2022.
- [23] K. Tan, W. Huang, X. Liu, J. Hu, and S. Dong, "A hierarchical graph convolution network for representation learning of gene expression data," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 8, pp. 3219–3229, Aug. 2021.
- [24] J. Xie et al., "Tracing unknown tumor origins with a biological-pathway-based transformer model," *Cell Rep. Methods*, vol. 4, no. 6, 2024, Art. no. 100797.
- [25] A. Khan and B. Lee, "DeepGene Transformer: Transformer for the gene expression-based classification of cancer subtypes," *Expert Syst. With Appl.*, vol. 226, 2023, Art. no. 120047.
- [26] M. Gokhale, S. K. Mohanty, and A. Ojha, "GeneViT: Gene vision transformer with improved DeepInsight for cancer classification," *Comput. Biol. Med.*, vol. 155, 2023, Art. no. 106643.
- [27] Y. Zhang and Q. Yang, "An overview of multi-task learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 30–43, 2018.
- [28] Z. Chen, X. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5177–5186.
- [29] K. G. Dizaji, W. Chen, and H. Huang, "Deep large-scale multitask learning network for gene expression inference," *J. Comput. Biol.*, vol. 28, no. 5, pp. 485–500, 2021.

- [30] G. He, M. Chen, Y. Bian, and E. Yang, "Mtm: A multi-task learning framework to predict individualized tissue gene expression profiles," *Bioinformatics*, vol. 39, no. 6, 2023, Art. no. btad363.
- [31] Z. Wang, Z. He, M. Shah, T. Zhang, D. Fan, and W. Zhang, "Network-based multi-task learning models for biomarker selection and cancer outcome prediction," *Bioinformatics*, vol. 36, no. 6, pp. 1814–1822, 2020.
- [32] D. Szklarczyk et al., "String V11: Protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets," *Nucleic Acids Res.*, vol. 47, no. D1, pp. D607–D613, 2019.
- [33] R. Ietswaart, B. M. Gyori, J. A. Bachman, P. K. Sorger, and L. S. Churchman, "GeneWalk identifies relevant gene functions for a biological context using network representation learning," *Genome Biol.*, vol. 22, pp. 1–35, 2021.
- [34] J. L. Katzman, U. Shaham, A. Cloninger, J. Bates, T. Jiang, and Y. Kluger, "DeepSurv: Personalized treatment recommender system using a Cox proportional hazards deep neural network," *BMC Med. Res. Methodol.*, vol. 18, no. 1, pp. 1–12, 2018.
- [35] X. Wu, Y. Shi, M. Wang, and A. Li, "CAMR: Cross-aligned multimodal representation learning for cancer survival prediction," *Bioinformatics*, vol. 39, no. 1, 2023, Art. no. btad025.
- [36] A. G. Robertson et al., "Comprehensive molecular characterization of muscle-invasive bladder cancer," *Cell*, vol. 171, no. 3, pp. 540–556, 2017.
- [37] C. G. A. R. Netw., "Comprehensive, integrative genomic analysis of diffuse lower-grade gliomas," *New England J. Med.*, vol. 372, no. 26, pp. 2481–2498, 2015.
- [38] D. A. Levine et al., "Integrated genomic characterization of endometrial carcinoma," *Nature*, vol. 497, no. 7447, pp. 67–73, 2013.
- [39] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.
- [40] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Representations*, 2015, pp. 1–14.
- [41] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [42] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 4700–4708.
- [43] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-ResNet and the impact of residual connections on learning," in *Proc. AAAI Conf. Artif. Intell.*, 2017, vol. 31, no. 1, pp. 4278–4284.
- [44] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 1251–1258.
- [45] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, vol. 30, pp. 5998–6008.
- [46] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017.
- [47] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Representations*, 2018.
- [48] M. Mostavi, Y.-C. Chiu, Y. Huang, and Y. Chen, "Convolutional neural network models for cancer type prediction based on gene expression," *BMC Med. Genomic.*, vol. 13, pp. 1–13, 2020.